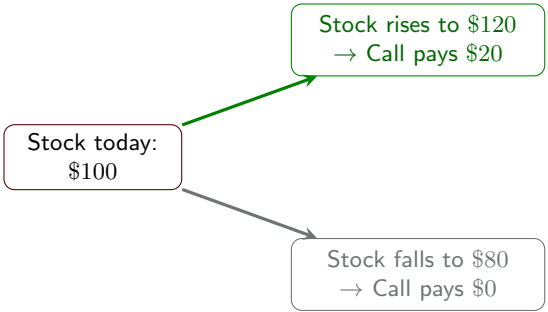
What Is an Option?

An option gives you the right, but not the obligation, to buy or sell an asset at a predetermined price.

- **Call:** right to buy at the strike price K.
- **Put:** right to sell at the strike price K.
- **Call payoff:** $\max(S_T - K, 0)$.
- **Put payoff:** $\max(K - S_T, 0)$.
- **Expiration T:** last date the right can be exercised.
- **Premium:** price paid today for the contract.

Pricing examples will focus on European call options, which can be exercised only at expiration.

A Stock with Two Possible Futures



Example: $100 strike price, payoff measured at expiration.

We know what the option pays in each scenario. But what should we pay for it today?

Today's option price = ?

Replicating the Financial Claim

Rather than price the claim directly, build a portfolio with Stocks and Loans.

Hold Δ shares of stock and borrow y dollars today at continuously compounded rate r.
Δ is the shares you have bought; y is the amount of the loan.

$$V_u = \Delta S_u - ye^{rT},$$
$$V_d = \Delta S_d - ye^{rT}.$$

Solving for Δ and the Loan

Subtract the down-state payoff from the up-state payoff:

$$\begin{aligned} V_u - V_d &= (\Delta S_u - ye^{rT}) - (\Delta S_d - ye^{rT}) \\ &= \Delta S_u - \Delta S_d \\ &= \Delta(S_u - S_d). \end{aligned}$$

$$\Delta = \frac{V_u - V_d}{S_u - S_d}$$

Then use either state, for example the down state:

$$\begin{aligned} V_d &= \Delta S_d - ye^{rT} \\ ye^{rT} &= \Delta S_d - V_d \\ y &= e^{-rT}(\Delta S_d - V_d). \end{aligned}$$

Δ is the shares you have bought; y is the amount of the loan.

No-Arbitrage:

Portfolio lemma

If two portfolios have identical future payoffs in every state, then an arbitrage-free market must assign them the same price today.

$$V_0 = \Delta S_0 - y$$

A One-Step Call Option Example

$S_0 = \$100$, $S_u = \$120$, $S_d = \$80$, $K = \$100$, and $r = 0$. Therefore,

$$V_u = \max(120 - 100, 0) = \$20, \qquad V_d = \max(80 - 100, 0) = \$0,$$

$$\Delta = \frac{20 - 0}{120 - 80} = 0.5, \qquad y = 0.5(80) - 0 = \$40,$$

$$V_0 = 0.5(100) - 40 = \boxed{\$10}.$$

Up state

Half a share is worth $\$60$. Repay $\$40$.
Remaining payoff: $\$20$.

Down state

Half a share is worth $\$40$. Repay $\$40$.
Remaining payoff: $\$0$.

The $\$10$ premium is paid today; $\$20$ or $\$0$ is paid at expiration.

Substituting the Replicating Portfolio

Start with the cost of the stock-and-loan portfolio:

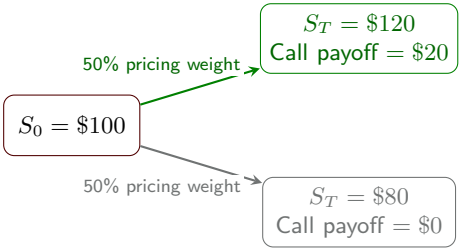
$$\begin{aligned} V_0 &= \Delta S_0 - y \\ &= \Delta S_0 - e^{-rT}(\Delta S_d - V_d) \\ &= e^{-rT}\left[\Delta(S_0 e^{rT} - S_d) + V_d\right]. \end{aligned}$$

Now substitute $\Delta = \dfrac{V_u - V_d}{S_u - S_d}$:

$$\begin{aligned} V_0 &= e^{-rT}\left[\frac{S_0 e^{rT} - S_d}{S_u - S_d}(V_u - V_d) + V_d\right] \\ &= e^{-rT}\left[qV_u + (1-q)V_d\right]. \end{aligned}$$

The stock-and-loan replica leads directly to a discounted, pricing-weighted payoff.

Pricing an Option Using Expected Payoffs



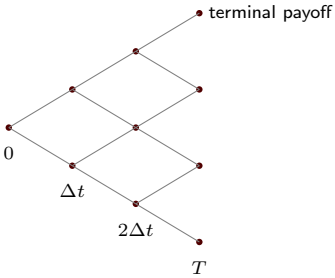
$$0.5(\$20) + 0.5(\$0) = \$10$$

These are risk-neutral pricing weights, not predictions of the stock's actual chances of rising or falling.

$$V_0 = e^{-rT}[qV_u + (1-q)V_d]$$

Expected future payoff, discounted to today.

Extending to Multiple Time Steps



At every terminal node, the contract tells us the payoff. Then work backward:

$$V_t = e^{-r\Delta t}\left[qV^u_{t+\Delta t} + (1-q)V^d_{t+\Delta t}\right].$$

1. Calculate terminal payoffs.
2. Weight and discount adjacent values.
3. Repeat until time zero.

Terminal nodes are model-generated scenarios, not forecasts of realized prices.

From Binomial Trees to Black–Scholes



When we make the time steps smaller and scale the possible movements correctly, binomial option prices approach Black–Scholes prices.

How Black–Scholes Calculates an Option Price

1. Model possible future stock prices $\longrightarrow$ 2. Calculate their option payoffs $\longrightarrow$ 3. Weight and discount to today

$$C_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)]$$

Expected call payoff, using risk-neutral probabilities, discounted to today.

The resulting Black–Scholes call price

$$\boxed{C_0 = S_0 N(d_1) - K e^{-rT} N(d_2)}$$

The Greeks: Sensitivities of Option Value

Delta: $\frac{\partial V}{\partial S}$

Sensitivity to stock price; locally, the shares needed to hedge the option.

Gamma: $\frac{\partial^2 V}{\partial S^2}$

Sensitivity of Delta to the stock price.

Theta

Sensitivity to time; its sign depends on whether time is calendar time or time to expiration.

Vega: $\frac{\partial V}{\partial \sigma}$

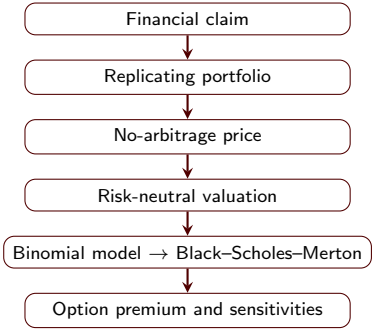
Sensitivity to volatility.

Rho: $\frac{\partial V}{\partial r}$

Sensitivity to the interest rate.

Connection: the Δ from the one-step replicating portfolio becomes a local, continuously adjusted hedge ratio.

Final Takeaway



Central message

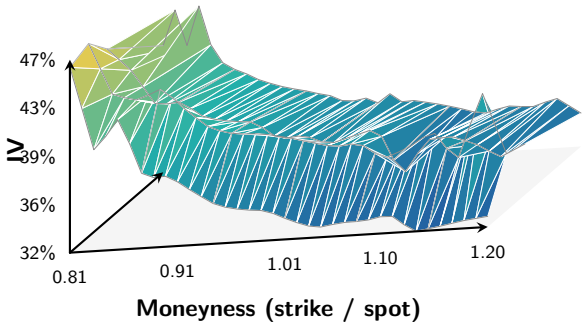
Options pricing is not fundamentally about predicting where a stock will go. It is about determining what a financial claim should cost today, given its possible future payoffs and the prices of tradable assets.

Practical Applications: The Options Chain Speaks

Practical Applications: Reading an Options Surface

NVDA implied volatility surface

Oct. 14, 2026 — x: moneyness — depth: expiration (DTE) — height/color: IV



Filtered to 0.80–1.20 moneyness and IV below 150%; interpolated.

Downside skew

At 0.90 × spot, IV is 3.6–4.5 points above the 1.10 × level. Lower strikes therefore carry higher implied volatility: a downside skew.

Across expirations

At-the-money IV: 33.8% (Nov. 13, 30d), 38.4% (Nov. 20, 37d), 37.2% (Nov. 27, 44d), and 36.6% (Dec. 18, 65d).

Adds missing repos and refreshes history, skipping Options updater, stocks update, and Options Model.

Refresh configured active repos

Run checkup for all teams

Repositories

Add repositories, open one to manage its team, refresh commit history, and inspect cached activity.

Grades cached GitHub activity from the last 7 days for every listed repository team.

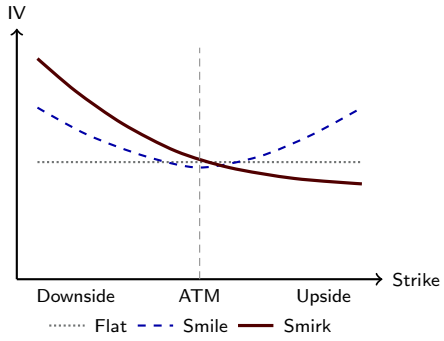
org

repo

Add repo

AggieInvestmentClub/AIC-BSM active	0 members	0 commits	RED 0/100	No cached commits	Open checkup Markdown PDF
AggieInvestmentClub/Factor-Tree-R-D active	2 members	2 commits	RED 19/100	Latest 2026-10-13 22:12:20	Open checkup Markdown PDF
AggieInvestmentClub/quant-fund-options-ml active	0 members	0 commits	RED 0/100	No cached commits	Open checkup Markdown PDF
AggieInvestmentClub/sellingputsQuantFund active	3 members	5 commits	RED 28/100	Latest 2026-10-13 17:05:52	Open checkup Markdown PDF

The Volatility Smirk: Why Downside Skew Exists



Same expiration: implied volatility across strikes.

An asymmetric curve

Flat skew assigns similar IV across strikes. A smile raises IV in both tails. A **smirk** is steeper on one side of the distribution than the other.

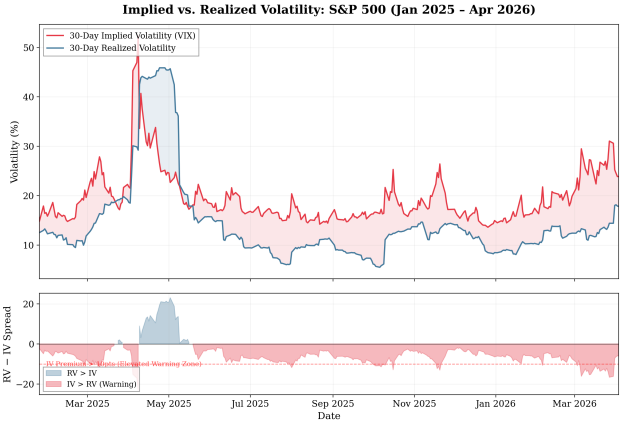
What downside smirk conveys

Market prices assign a larger effect on implied volatility to a downside move than to a comparable upside move. Lower-strike options therefore trade at higher IV.

How does skew exist? Supply meets demand

Investors buy puts to protect portfolios against falling markets, which have historically produced larger and more severe moves. That demand raises downside-option prices and their implied volatilities. At a fundamental level, skew emerges as buyers and sellers meet at different prices across strikes; implied volatility is simply those option prices expressed in volatility terms.

Why Implied Volatility Often Exceeds Realized Volatility



S&P 500, Jan. 2025–Apr. 2026. A negative IV spread means implied volatility exceeded trailing realized volatility.

Demand for options

Traders pay a premium for efficient, leveraged exposure to the underlying asset and, more importantly, protection against adverse price movements.

Compensation for sellers

Option sellers are the counterparties to that demand. They require compensation for bearing volatility risk and the possibility of extreme tail losses.

The volatility risk premium

Together, option demand and seller compensation contribute to implied volatility often exceeding subsequent realized volatility. Academics call this difference the **volatility risk premium**.

Iron Condor: Strike Math and Payoff



SPY example shown: $771.77 spot, 22 DTE, 12.9% implied volatility.

The short-volatility trade

Investors attempt to capture the volatility risk premium by selling options and accepting exposure to volatility and potentially severe market movements. The strategy can generate consistent premiums during relatively stable markets, but may experience substantial losses during periods of extreme volatility.

Four legs, one credit

Buy 762P / sell 767P; sell 777C / buy 782C. Both wings are $5 wide. Net credit and max profit: $3.78 per share, or **$378 per 1-lot**. Max loss: $122.

At expiration

Max profit occurs when SPY finishes between the two short strikes ($767–$777). Beyond either long wing, the loss is capped at $122 per 1-lot.

Appendix: Risk-Neutral Pricing Weight

$$q = \frac{S_0 e^{rT} - S_d}{S_u - S_d}$$

The pricing weights q and $1 - q$ make the discounted expected future stock price equal to today's stock price.